drylab instructor key — v1 (for exercises revision 3, drylab v0.24.1)

Every number below was **measured against drylab v0.24.1** on 2026-08-18; where a value depends on conditions, the operating point is stated. Slider values are exact UI settings, so this key doubles as a self-run script: follow the *Do* lines top to bottom and the screen should show the *Expect* values. Small deviations (last digit of Rs, ±1 transfer) are normal; a disagreement in sign, order, or factor-of-two is not — report it.

Conventions: default HPLC column is the Zorbax SB-C18 preset at **100 × 4.6 mm, 3 µm** (not the 250 × 5 of the physical product — worth knowing before checking numbers against intuition). GC default column is 30 m × 0.25 mm, d_f 0.25 µm, He, inlet 250 kPa → ambient outlet.

Exercise 1 — Craig apparatus

Do: Craig tab. D_A = 4, D_B = 0.25, both volumes 1.0, tubes ≥ 5. (The worksheet's advanced section sets tubes to 250 — available from drylab v0.24.3.)

- n = 2 → **Expect** per-tube fractions (tubes 0,1,2): D = 4 gives **0.04 / 0.32 / 0.64**; D = 0.25 mirrors it: **0.64 / 0.32 / 0.04**. Hand check:

$$p = \frac{D V_u}{D V_u + V_l} \overset{V_u = V_l}{=} \frac{D}{D+1} = 0.8 \text{ and } 0.2, \qquad (q^2,\ 2pq,\ p^2)$$

- Transfers 5 → 25 → 100: centers at n·p separate linearly; widths grow as √(npq). At n = 100 the D = 4 band centers near tube 80, D = 0.25 near tube 20.
- Normal overlay at n = 100: matches closely; the Gaussian is the binomial's limit — each transfer is one equilibration = one plate.

Quantitative answers:

- D = 1, equal volumes, n = 50: r_max = n·p = **25**, fraction there **0.112**. D = 3 with V_upper = 2·V_lower:

$$p = \frac{D V_u}{D V_u + V_l} = \frac{3 \cdot 2}{3 \cdot 2 + 1} = \frac{6}{7} = 0.857 \ \Rightarrow\ r_{max} = 50\,p \approx 43$$

The trap is using D/(D+1) = 0.75 → tube 37–38; the volume-weighted p is the lesson.

- D = 1 vs D = 4 (p = 0.5, 0.8): Rs ≈ 1.5 when

$$n\,\Delta p \ \geq\ 3\,(\sigma_1 + \sigma_2), \qquad 0.3\,n \ \geq\ 2.7\sqrt{n} \ \Rightarrow\ n \approx 81$$

the slider search should land in the 75–90 range depending on how the student reads "resolved."

Advanced section (worksheet “Going further”, added 2026-08-26; requires drylab ≥ 0.24.3 for the 250-tube axis):

- **A1 — scaling law.**

$$R_s = \frac{n\,|\Delta p|}{2(\sqrt{n\,p_A q_A} + \sqrt{n\,p_B q_B})} = \frac{|\Delta p|\,\sqrt{n}}{2(\sqrt{p_A q_A} + \sqrt{p_B q_B})} \overset{p = 0.5,\ 0.8}{=} \frac{0.3\sqrt{n}}{2\,(0.5 + 0.4)} = \frac{\sqrt{n}}{6}$$

Panel readouts: **1.18 (n=50), 1.67 (n=100), 2.36 (n=200)** — exact to the displayed digits; each doubling multiplies Rs by **√2 = 1.414**.

- **A2 — placement at fixed selectivity** (D_B/D_A = 4, n = 100):

(D_A, D_B)	(0.125, 0.5)	(0.25, 1)	(0.5, 2)	(1, 4)	(2, 8)	(4, 16)
Rs	1.41	1.67	**1.77**	1.67	1.41	1.11

The ridge is symmetric with its maximum at **(0.5, 2)** — geometric mean of the D’s at 1, i.e. the p’s straddling ½ (⅓ and ⅔). The trade: large D’s compress both p’s toward 1 (Δp collapses faster than the widths shrink); small D’s mirror it. This is the Craig-machine form of the retention *window* — penalties on both sides — that reappears as the k/(1+k) saturation in the HPLC triage exercise.

- **A3 — volume rescue.**

$$p = \frac{D\,V_u}{D\,V_u + V_l} = \frac{D\,(V_u/V_l)}{D\,(V_u/V_l) + 1}$$

so the effective distribution ratio is D·(V_u/V_l). Moving (4, 16) onto the A2 optimum (0.5, 2) needs **V_u/V_l = 1/8** (set upper 1, lower 8 — or 0.125 and 1). Verified: Rs recovers to **1.77** at n = 100, the ridge maximum exactly. Column bridge: the volume ratio is the **phase ratio β** ($k' = K/\beta$), moved in a real column by film thickness / bore (GC) or bonded-phase loading and geometry (LC). Accept any consistent pair of volumes with the derivation shown.

Exercise 2A — van Deemter term builder

Do: Band shape → Mechanisms. Defaults: d_p 3 µm, λ 1.0, D_m 1.0, C_s 2 ms, C_m 1 ms.

- **Expect at defaults:** A = 6.0 µm, optimum stamp reads **u_opt = 0.063 cm/s, H_min = 9.79 µm**.
- C_s = C_m = 0 → only the flat A line and the falling B/u limb remain; stamp reads “no optimum (C = 0)”.
- Slider table (one change from defaults each time):

Change	A (µm)	u_opt (cm/s)	H_min (µm)	Reading
defaults	6.0	0.063	9.79	—
d_p 3 → 6	**12.0**	0.063	15.79	A = 2λd_p; C sliders don’t move (they’re entered directly — that’s the discussion hook)

Change	A (µm)	u_opt (cm/s)	H_min (µm)	Reading
λ 1.0 → 0.5	**3.0**	0.063	6.79	better packing buys A only
D_m 1 → 2	6.0	**0.089**	11.37	B doubles; u_opt scales as √D_m
C_s 2 → 4	6.0	**0.049**	10.90	steeper fast limb, optimum retreats
C_m 1 → 0	6.0	0.077	9.10	flatter fast side

Conceptual answers: open-tubular columns have no A term because there is no packing (one flow path). The C expressions students should write: C_s ∝ d_f²/D_s, C_m ∝ d_p²/D_m; in the HPLC section the particle-size slider moves C_m (and A), temperature moves D_m and thus B and C_m. "Same H_min, different shape → run the flatter one fast; the number is the C sum."

Quantitative: from the Term curves CSV: A = the constant column (6.0 µm); B = (b_um · u) at any row (= 1.2×10^{-5} cm²/s → 2γD_m ✓); C = slope of (cs+cm) vs u (= 3 ms). Then

$$u_{opt} = \sqrt{B/C}, \qquad H_{min} = A + 2\sqrt{B\,C}$$

must reproduce the stamp — exact by construction.

Exercise 2 — van Deemter on the HPLC model

Do: HPLC, Default preset. Sweeps tab: flow_rate → hetp.

- **Expect:** minimum at **≈ 1.6 mL/min, h_min ≈ 6.0 µm** (= 2·d_p); h ≈ 15.4 µm at 0.2 mL/min, ≈ 8.1 µm at 5.0. Operating point on the plot: 100×4.6 mm 3 µm, 50 %B MeCN, 25 °C.
- Temperature probe: at 70 °C the minimum shifts right; at 10 °C left (through D_m — B up and C down with T).
- Honest framing for "why not run at optimum": at 3 µm the default 2.0 mL/min *is* near-optimal; the classic run-fast tradeoff appears at 6 µm or low T.

Quantitative (fit of the exported sweep CSV, h in µm, F in mL/min):

$$h(F) = A + \frac{B}{F} + C\,F, \qquad F_{opt} = \sqrt{B/C}, \qquad h_{min} = A + 2\sqrt{B\,C}$$

- 3 µm: **A = 3.00, B = 2.44, C = 0.922** → F_opt = 1.63, h_min = 6.00 ✓.
- 6 µm: **A = 6.00 (×2), B = 2.44 (unchanged), C = 3.69 (×4)** → F_opt = 0.81, h_min = 12.0. Backpressure at 2 mL/min: **173 bar (3 µm) vs 43 bar (6 µm)** — the ΔP ∝ 1/d_p² sentence writes itself.
- (If students fit in velocity units instead of flow, coefficients rescale but F_opt·(conversion) and h_min must agree — either is acceptable if they say which they did.)

Exercise 3 — Resolution triage

Do: HPLC preset **Ex 3** (`?hplc=ex3`). Results tab.

- **Expect on open:** acetophenone then p-chlorophenol, **Rs = 0.65, α = 1.031**, N ≈ 16 500, k′₂ = 2.08, backpressure 173 bar, t_{R2} ≈ 98 s.
- Team outcomes (one lever at a time from the preset):

Team	Move	Expect
k	50 → 44 %B	**Rs 2.58** (α also rises to 1.115 — say so up front)
α	MeOH, 65 %B	**Rs 7.79, α 1.57** — chemistry wins
N	300 mm	Rs 1.14, 520 bar, t_{R2} 295 s
N	300 mm **and** 1.5 µm	**Rs 1.56 at 2082 bar** — technically fixed, physically absurd

Quantitative answers: fundamental equation with the table's N, α, k′₂:

$$R_s = \frac{\sqrt{N}}{4} \cdot \frac{\alpha - 1}{\alpha} \cdot \frac{k'_2}{1 + k'_2} = \frac{\sqrt{16521}}{4} \cdot \frac{0.031}{1.031} \cdot \frac{2.082}{3.082} = \mathbf{0.654}$$

which matches the table's Rs to three figures at this preset (both widths are the model's, and extra-column terms are small here; if a student's numbers disagree, the usual cause is using k′₁ or unsorted rows). Length prediction: 0.65·√3 = 1.13 vs measured **1.14** ✓, at 3× time and 3× pressure — tripling the column is the right *first* move approximately never.

Exercise 4 — General elution problem

LC leg. Do: preset **Ex 4** (`?hplc=ex4`, all 22 compounds, 30 %B).

- **Expect:** k′ spans **1.28 → 352**, last peak ≈ **188 min**. At 70 %B: k′ 0.22 → 5, everything inside **3 min**, early crowd. The student gradient should land every k′ in roughly 1–10; the Band-tracks view shows each k′ collapsing as the front arrives.
- Dwell forensics: defaults 200 + 200 µL at 2 mL/min → expected delay

$$t_{dwell} = \frac{V_{dwell}}{F} = \frac{400\ \mu L}{2\ \text{mL/min}} = 12\ \text{s}$$

visible as the offset between the programmed and delivered profiles in the Solvent-profile view.

GC leg. Do: preset **Ex 4** (`?gc=ex4`, C7–C15 on SLB-5ms), 120 °C; then Programmed tab with the default program (40 °C, 1 min hold, →300 °C by 27 min); Compare tab.

- **Expect isothermal:** 9 peaks, 87 s → 1047 s (17.4 min), early alkanes crowded, C15 flat.
- **Expect programmed:** peaks 139 s → 869 s with near-constant widths (mean τ ≈ 0.5 s).
- Peak capacity from the exported results:

$$n_c^{iso} = 1 + \frac{\sqrt{\bar{N}}}{4} \ln \frac{t_{last}}{t_{first}} \approx \mathbf{259}, \qquad n_c^{prog} = 1 + \frac{t_{last} - t_{first}}{4\,\bar{\tau}} \approx \mathbf{381}$$

→ ratio ≈ **1.5×**, and the qualitative win (late peaks sharp, run 3 min shorter) is bigger than the ratio alone suggests. Accept ±10 % — the formulas are estimates and students may average N differently.

Exercise 5 — GC retention: polarity and thermodynamics

Do: presets ex5a → ex5b → (phase to Wax) → ex5c.

- **Ladder (ex5a):** Octane < Decane < Dodecane at 120 °C; k roughly multiplies per 2 CH_2 (read k from the table).
- **Polarity trio (ex5b):** on SLB-5ms **2-Octanone < Decane < 1-Octanol**; switch phase to Wax → **Decane < 2-Octanone < 1-Octanol**. The alkane vaults to first; the alcohol anchors last on both.
- **Golay coda** (Decane, SLB-5ms, 120 °C, ambient outlet): **H_min is the same for all three gases, 0.137 mm** — the Golay minimum depends on geometry and k, not the carrier; the *carrier* sets where and how forgiving it is: u_opt ≈ **20 (N_2), 48 (He), 69 (H_2) cm/s**, and at the pressure ceiling H ≈ 0.86 (N_2) vs 0.25 (He) vs 0.42 mm (H_2, at 210 cm/s). "Which to buy": H_2 runs 3× faster than N_2 at equal H.

Quantitative (Wax, 4 points at 100/120/140/160 °C, k from the table):

Solute	k @ 100/120/140/160 °C	slope (K)	−ΔH° (kJ/mol)
Decane	0.339 / 0.202 / 0.130 / 0.088	3623	**30.1**
1-Octanol	8.055 / 3.451 / 1.652 / 0.869	6001	**49.9**
Tridecane	1.868 / 0.932 / 0.511 / 0.303	4901	40.8
2-Octanone	1.862 / 0.963 / 0.542 / 0.328	4680	38.9

The fit model and its reading:

$$\ln k = -\frac{\Delta H^\circ}{R} \cdot \frac{1}{T} + \text{const} \;\Rightarrow\; \Delta H^\circ = -R \times \text{slope}$$

ΔH ranking (alcohol ≫ alkane on Wax) matches the Round-2 order argument. **Co-elution (ex5c):** intersection of the two 4-point fits predicts **≈ 101 °C** (exact model crossing ≈ 102 °C; the fit is over a curved line, so ±2 °C is success). Walking the oven slider down from 120 °C, the pair merges near 100–105 °C and inverts below.

Exercise 6 — CE

Do: CE tab, defaults (30 mm, 500 V, detector 15 mm, run 20 s, K^+/Na^+/Li^+ over acetate/Tris). Press Run.

- **EOF = 0, expect peaks at ≈ 8.3 s (K^+), 11.7 s (Na^+), 14.7 s (Li^+)** — crystal-radius order inverted by hydration. Mobility arithmetic: the plug is centered at **6 mm**, so use 9 mm to the detector:

$$t = \frac{L_{det}}{\mu_{app} E}, \qquad \mu_{app} = \mu_{ep} + \mu_{EOF}, \qquad E = \frac{500\ \text{V}}{30\ \text{mm}} = 16.7\ \text{kV/m}$$

giving 7.1 / 10.4 / 13.5 s for the 9 mm path — order and ratios right, simulated times ~15 % longer because effective mobilities in the real electrolyte are below the infinite-dilution table values. A student who reconciles order + ratio + the systematic has fully succeeded.

- Acetate: **in the default composition acetate is BGE — it fills the capillary and shows no peak.** The first (intended) stumble is realizing this and switching its role to Sample in the composition editor. Then, with the plug at 6 mm (9 mm to the detector): at EOF 60, μ_app = +17.6 → **≈ 31 s — outside the 20 s default run**; within 20 s needs μ_app ≥ 27, i.e. **EOF ≥ ≈ 70**; within a 60 s run needs only μ_app ≥ 9, i.e. **EOF ≥ ≈ 52** (bracket at 50–55 on the slider). Any consistent pair of (EOF, run time) with the arithmetic shown is full credit. Expect simulated arrivals ~10–15 % later than the arithmetic, as with the cations.

- N vs V: run 500/1000/1500 V and compute, from the on-screen migration time and FWHM,

$$N = 5.54 \left(\frac{t}{w_{1/2}}\right)^2$$

  Expect roughly linear growth until the Joule report's ΔT becomes non-trivial at the top of the range; the break point depends on the composition, so grade the *method* (linear check + correlation with the Joule number), not a specific voltage.

Exercise 7 — Overload and asymmetry

Do: Band shape → Peak shape, defaults (N = 100, k = 2, curvature 0, load 1).

- **Baseline: t_max = 297, As = 1.18 for BOTH curves, N_app = 150** vs built-in 100 (the moment-based N of a negative binomial exceeds the plate number — a good early "what does this number assume?" moment). N = 1000: As_ideal falls to **1.05**.

- Curvature/tau grid (N = 100, k = 2, load 1 unless stated):

Setting	t_max	As	N_app	Reading
linear	297	1.18	150	baseline
χ = +1	291	1.29	147	tailing begins, max walks early
χ = +2	285	1.43	143	

Setting	t_max	As	N_app	Reading
χ = +5	**267**	**1.99**	126	strong Langmuir
χ = −0.4	300	1.11	151	fronting: As *below* baseline, max late
χ = −0.85	303	**1.06**	151	clamp limit; announced in a callout
τ = 15, χ = 0	**310**	1.26	**120**	kinetic tailing: max moves LATE
curv 0.5 × load 4	285	1.43	143	≡ χ = 2 — curvature×load is the coupling
curv 0.5 × load 10	267	1.99	126	≡ χ = 5

- Diagnostic answer: cut the load — overload As collapses toward 1.18, kinetic As doesn't; and overload moves t_max *early* while slow-site τ moves it *late*. Two observables, fully separating the mechanisms.
- QC design answer: the load ladder drops N_app 150 → 126 (−16 %) on an unchanged column; at slightly higher load the −20 % flag fires. The saving readouts: As rose and t_max moved *earlier* — a degraded column (kinetic tailing) shows As rising with t_max moving *later* and no load dependence.
- CSV answers (χ = +2): As from the exported points ≈ 1.43 ✓; the Langmuir tail decays toward the linear-isotherm retention (n ≈ 300) because the most dilute part of the band travels at the linear velocity — the tail *points at* the analytical tR; the front carries the isotherm (shock position moves with load).

If a number disagrees

Check, in order: (1) the preset actually loaded (the URL param spelling); (2) a control drifted from the stated operating point — every exported CSV carries its operating-point stamp for exactly this comparison; (3) the version (this key: v0.24.1). If all three check out, the disagreement is real and worth reporting — several numbers in this key exist because writing it caught errors in the exercises.